

A Real-Time Terrain-Adaptive Local Trajectory Planner for High-Speed Autonomous Off-Road Navigation on Deformable Terrains

Siyuan Yu^{ID}, Congkai Shen^{ID}, James Dallas^{ID}, Bogdan I. Epureanu^{ID}, Paramsothy Jayakumar, and Tulga Ersal^{ID}

Abstract—This paper presents a novel terrain-adaptive local trajectory planner designed for the autonomous operation of off-road vehicles on deformable terrains. State-of-the-art solutions either do not account for deformable terrains, or do not offer sufficient robustness or computational speed. To bridge this research gap, the paper introduces a novel model predictive control (MPC) formulation. In contrast to the prevailing state-of-the-art approaches that rely exclusively on hard or soft constraints for obstacle avoidance, the present formulation enhances robustness by incorporating both types of constraints. The effectiveness and robustness of the formulation are evaluated through extensive simulations, encompassing a wide range of randomized scenarios, and compared against state-of-the-art methods. Subsequently, the formulation is augmented with an optimal-control-oriented terramechanics model from the literature, explicitly addressing terrain deformation. Additionally, a terrain estimator employing the unscented Kalman filter is utilized to dynamically adjust the sinkage exponent online, resulting in a terrain-adaptive formulation. This formulation is tested on a physical vehicle in real world experiments against a rigid-terrain formulation as the benchmark. The results showcase the superior safety and performance achieved by the proposed formulation, underscoring the critical significance of integrating terramechanics knowledge into the planning process. Specifically, the proposed terrain-adaptive formulation achieves reduced mean absolute sideslip angle, decreased mean absolute yaw rate, shorter time to goal, and a higher success rate, primarily attributed to its enhanced understanding of terramechanics within the planner.

Index Terms—Local trajectory planning, model predictive control, collision avoidance, autonomous ground vehicles, physical vehicle testing, vehicle dynamics, terrain estimation, terramechanics, deformable terrains.

I. INTRODUCTION

AUTONOMOUS Ground Vehicles (AGVs) are poised to play crucial roles in off-road navigation applications,

including military operations, mining, construction, and planetary exploration. One of the paramount challenges they face is navigating through deformable terrain – a factor that profoundly influences vehicle dynamics and, consequently, mobility. Traversing deformable terrains is fraught with added complexities in contrast to rigid terrains or roads, since terrain deformations can potentially hinder or even obstruct vehicular movement completely. This challenge is exacerbated by the fact that the terrain’s properties are often unknown a priori and can even change dynamically in response to environmental factors like moisture content. Hence, to unlock advanced off-road navigation capabilities, it is not simply beneficial but imperative to understand how terrain deformations affect vehicle mobility and to estimate the terrain’s characteristics in real time and in situ. Off-road environments present additional uncertainties. They are unstructured, with a wide range of obstacles, and typically devoid of designated pathways or markings to indicate safe paths and speed limits.

Thus, for AGVs to attain higher levels of autonomy in off-road environments, navigation algorithms must capture the interplay between tires and deformable terrain to create dynamically feasible trajectories and maximize mobility, even when terrain properties change with environmental factors. Despite the absence of the structured indicators found in traditional road environments, AGVs must be capable of navigating safely by adapting to the unpredictable nature of deformable terrains.

However, as the literature review below reveals, existing navigation frameworks either lack a comprehension of terramechanics or have yet to deliver a solution that is both robust and computationally efficient.

A. Background

Several planning algorithms have been developed and evaluated for autonomous vehicle navigation. One frequently employed approach involves the utilization of graph-search methods. For instance, in 2D planning scenarios, the D* Lite algorithm [1], facilitates prompt replanning in response to unforeseen hazards. To mitigate the inherent discontinuity introduced from grids in such methods, researchers have devised smoothing techniques [2], [3]. When the dimensionality of the problem increases, sampling-based methods, such as rapidly exploring random trees (RRT), may be more suitable. For instance, in [4], a three-phase path planning

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Siyuan Yu, Congkai Shen, James Dallas, Bogdan I. Epureanu, and Tulga Ersal are with the Department of Mechanical Engineering, University of Michigan, Ann Arbor, MI 48109 USA (e-mail: johnnysy@umich.edu; cosh@umich.edu; dallasja@umich.edu; epureanu@umich.edu; tersal@umich.edu).

Paramsothy Jayakumar is with U.S. Army Ground Vehicle Systems Center, Warren, MI 48397 USA (e-mail: paramsothy.jayakumar.civ@army.mil).

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algorithm is developed that takes non-holonomic constraints and 3D terrain topology into account. However, the ability of these solutions to deliver satisfactory safety and performance within the off-road mobility context has not yet been demonstrated due to the absence of consideration for dynamical constraints.

Model predictive control (MPC) has been utilized to explicitly consider vehicle dynamics in planning by optimizing trajectories under both linear and nonlinear constraints. MPC has demonstrated its ability to navigate a car in on-road scenarios characterized by structured and bounded environments during ordinary driving tasks [5], [6]. In [7], the MPC formulation is specifically designed to address the challenge of avoiding imminent collisions at high speeds, where the vehicle is pushed to the edge of its operational capabilities, emphasizing the significance of considering vehicle dynamics and tire-road interactions. MPC has also been employed for obstacle avoidance maneuvers in unstructured, off-road circumstances, where there are no predetermined lane boundaries or paths [8]. Subsequently, these formulations were extended to address challenges related to moving obstacle avoidance, as well as vehicle planning and control for 3D terrains [9], [10]. However, these studies still neglect the terrain deformations.

Extending the MPC framework to deformable terrains requires a model of the tire-terrain interaction. Among the existing terramechanics models, the Bekker-based models have demonstrated sufficient accuracy in capturing steady-state operations [11], [12], [13]. The soil contact model (SCM) has advanced the modeling of tire-terrain interaction via discretization, thereby enabling application to dynamic operations, as well [14], [15]. However, SCM was found to not possess the real-time performance and twice continuous differentiability required for MPC. A recent study introduced a neural network-based terramechanics model trained using SCM, resolving the concerns regarding computational efficiency and the lack of twice continuous differentiability [16]. Subsequently, this neural network model was employed in conjunction with an MPC formulation for prediction and control, as well as an unscented Kalman filter (UKF) for adaptability [16]. Fidelity testing and MPC results collectively demonstrated the benefits of including this model for navigation on soft soils [16]. This effort made an MPC update rate of 2 Hz possible even when terrain deformations are taken into account. However, three important shortcomings are noted. First, an update rate higher than 2 Hz may be needed in certain applications, which the state-of-the-art does not yet meet. Second, past evaluations were performed in relatively simple environments with up to only two obstacles. Given that the MPC formulation included one hard constraint for each obstacle, it may not be feasible to maintain an update rate of at least 2 Hz in more complex environments, because as the number of constraints increases, the computational performance deteriorates. Third, robustness evaluations have been limited to the uncertainties in the terrain parameters estimated by the UKF.

In [17], [18], and [19], the controller is enhanced to account for the presence of deformable terrain using either

physics-based surrogate models or semi-physics-based surrogate models. Gupta et al. [20] employs reinforcement learning to control vehicles on deformable terrain. However, all these studies demonstrate their effectiveness exclusively in simulation environments, leaving real-world applications unexplored.

B. Original Contributions

Although previous research demonstrated successful utilization of MPC in both on- and off-road scenarios, high-speed autonomous navigation on deformable terrains still poses unique challenges: 1) While the literature did consider deformations and uncertainties in the terrain [16], the MPC formulation needs to be both scalable and robust to effectively handle complex environments. 2) Timely trajectory planning becomes critical at high speeds, where the available reaction time is significantly reduced.

This study tackles these challenges through a novel MPC formulation that operates at high update rates (10 Hz or higher), effectively scales to environments with hundreds of obstacles, and demonstrates robustness against a wide range of uncertainties. To the authors' best knowledge, this is the first experimentally validated algorithm that can push an autonomous off-road vehicle to its dynamic limits on deformable terrains. The prediction model in terms of vehicle dynamics and terramechanics, and the UKF filter for adaptation to unknown terrains follow those of [16]. Specifically, to capture the interaction of the tire with deformable terrain, a neural network based terramechanics model is utilized in the vehicle dynamics model for prediction. Building upon it, the MPC is augmented with a terrain estimator to track the changes in terrain parameters in real time and adapt the planned trajectory accordingly. Unlike [16], however, the MPC is formulated and solved with the following characteristics to enable a high update rate and robustness: 1) obstacle avoidance is enforced through a combination of hard and soft constraints instead of hard constraints alone; 2) an option to aggregate the obstacle avoidance constraints is included to maintain real-time performance in cluttered environments instead of having a separate constraint for each obstacle; 3) a fixed time horizon is used instead of considering time as a design variable; and 4) the backward Euler collocation approach is used instead of Legendre-Gauss-Radau (LGR). To address the challenge of navigating off-road environments without designated pathways or markings to indicate safe paths and speed limits, the MPC is designed to plan dynamically feasible and collision-free trajectories without the help of pre-defined trajectories.

The work goes even further beyond the state of the art by subjecting the algorithm to comprehensive testing in simulation under various sources of uncertainty, and experimental validation in a real-world environment.

The original contributions are thus summarized as follows:

- 1) A real-time MPC formulation with significantly higher update rate than the state of the art to plan collision-free and dynamically feasible trajectories on deformable terrains with nonlinear vehicle dynamics.

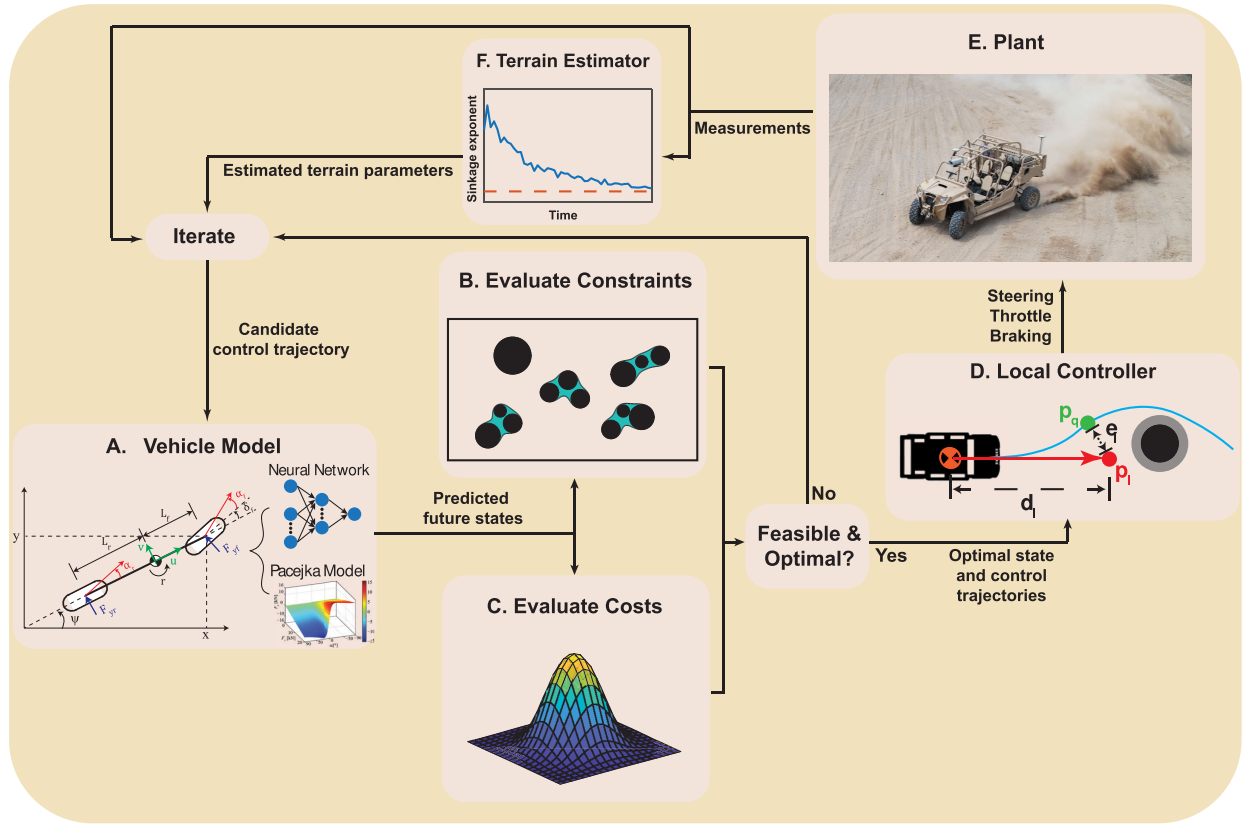


Fig. 1. The proposed framework employs Terrain-Adaptive Model Predictive Control (MPC) as a planner and a local controller to transmit commands to the actual vehicle. (A) Utilizing a 3 Degree of Freedom (DoF) bicycle model in conjunction with a neural network tire model for deformable terrains or a Pacejka tire model for rigid terrains, as detailed in Sec. II-A. (B) Demonstrating Kreisselmeier-Steinhauser (KS) constraint aggregation for obstacle avoidance, as expounded in Sec. II-D.2. The black circles represent obstacles, while the cyan areas denote conservative infeasible regions determined by KS constraint aggregation. (C) Presenting a surface plot of the potential field for obstacle avoidance, elucidated in Sec. II-D.3. (D) Depicting the diagram for the path tracker, as detailed in Sec. II-E. The blue curve illustrates the planned path. p_l represents the look-ahead point, while p_q signifies the queried point closest to the look-ahead point on the planned path. The look-ahead point is determined by extending a straight line from the center of mass along the heading with a look-ahead distance d_l . (E) Providing a snapshot of the MRZR operating autonomously on the testing field, outlined in Sec. II-F. (F) Illustrative example of terrain parameter estimation, described in Sec. II-C. The blue solid line represents the estimation of the sinkage exponent, while the red dashed line denotes the ground truth.

- 2) A validation of performance robustness under various sources of uncertainty through Monte Carlo simulations.
- 3) Real-world testing and validation of the proposed Terrain-Aware and Terrain-Adaptive MPC frameworks against the state-of-the-art benchmarks.

The remainder of the paper is organized as follows. Sec. II covers the vehicle dynamics along with the various representations of tire forces, the suggested MPC formulation, and the planning and control framework. Sec. III presents the robustness test of the MPC formulation. Sec. IV discusses the experimental results with the Terrain-Aware MPC formulation and further describes the enhanced performance achieved through the adoption of the terrain-adaptation feature. Finally, Sec. V provides concluding remarks.

II. METHODS

A. Vehicle Dynamics

A 3 Degree-of-Freedom (DoF) bicycle model is utilized as the prediction model for the vehicle dynamics and is

displayed in Fig. 1-A. It has been demonstrated in earlier research that this model maintains good fidelity when subjected to high lateral forces [21]. It thereby achieves a balance between computational performance and model fidelity.

The definitions of the state and control vectors are

$$\xi := \begin{bmatrix} x \\ y \\ \psi \\ u_x \\ v \\ r \\ \delta_f \\ a_x \end{bmatrix} = \begin{bmatrix} \text{global x position of front axle} \\ \text{global y position of front axle} \\ \text{yaw angle} \\ \text{longitudinal speed} \\ \text{lateral speed} \\ \text{yaw rate} \\ \text{front tire steering angle} \\ \text{longitudinal acceleration} \end{bmatrix} \quad (1)$$

$$\zeta := \begin{bmatrix} \dot{\delta}_f \\ \dot{j}_x \end{bmatrix} = \begin{bmatrix} \text{front tire steering rate} \\ \text{longitudinal jerk} \end{bmatrix} \quad (2)$$

The state-space representation for this vehicle model is

$$\dot{\xi} = A(\xi) + B\zeta \quad (3)$$

where

$$A(\xi) = \begin{bmatrix} u_x(t) \cos(\psi(t)) - (v(t) + L_f r(t)) \sin(\psi(t)) \\ u_x(t) \sin(\psi(t)) + (v(t) + L_f r(t)) \cos(\psi(t)) \\ r(t) \\ a_x(t) \\ (F_{yf} + F_{yr})/M - u - x(t)r(t) \\ (F_{yf}L_f - F_{yr}L_r)/I_{zz} \\ 0 \\ 0 \end{bmatrix} \quad (4)$$

$$B^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

Here M is the vehicle total mass, I_{zz} is the yaw moment of inertia, and L_f and L_r are distances from the vehicle center of gravity to the vehicle front and rear axles, respectively. F_{yf} and F_{yr} are vehicle front and rear tire forces. Two tire force models are utilized in the prediction models to allow for comparisons with the state of the art. Both models take the tire slip angles as the inputs, which are defined as

$$\alpha_f = \text{atan}\left(\frac{v + L_f r}{u_x}\right) - \delta_f \quad (6)$$

$$\alpha_r = \text{atan}\left(\frac{v - L_r r}{u_x}\right) \quad (7)$$

1) *Pacejka Tire Force*: To emphasize the importance of tire-terrain interactions, a state-of-the-art benchmark is set using the standard pure-slip Pacejka magic formula [22], representing a state-of-the-art tire model for on-road applications. The lateral forces in Eq. (4) are expanded as:

$$\begin{aligned} F_{y\bullet} &= D_\bullet \sin\left(C_\bullet \tan^{-1}\left(X_\bullet - E_\bullet X_\bullet + E_\bullet \tan^{-1}(X_\bullet)\right)\right) \\ X_\bullet &= B_\bullet \alpha_\bullet \end{aligned} \quad (8)$$

Here $\bullet = \{f, r\}$ stands for subscripts for front and rear, respectively. $B_\bullet$, $C_\bullet$, $D_\bullet$ and $E_\bullet$ are all functions related to the vertical load $F_{z\bullet}$. The Pacejka model is parameterized from on-road experiments for a Polaris MRZR 4 and provided by the U.S. Army Ground Vehicle Systems Center.

2) *Neural Network Terramechanics Model*: A neural network based terramechanics model is described in Sec. II-B. With this model, two requirements are satisfied:

- 1) twice continuous differentiability, which is key to maintaining high efficiency in solvers [23], and
- 2) high-fidelity prediction of lateral tire forces on deformable terrains, which is indispensable for dynamically feasible trajectory planning on such terrains, especially at high speeds.

The neural network terramechanics model is expressed as:

$$F_{y\bullet} = \text{NN}_{\text{tm}}(F_{z\bullet}, \alpha_\bullet, \kappa_\bullet, \dot{\delta}_\bullet, u_{w\bullet}, \mathcal{P}) \quad (9)$$

where $\bullet = \{f, r\}$ stands for either front or rear axles, $\kappa_\bullet$ is the vehicle slip ratio, and $\dot{\delta}_\bullet$ is the steering rate. The rear axle does not have an active steering system, resulting in a constant value of 0 for $\dot{\delta}_r$. The wheel rotation speed is denoted as $u_{w\bullet}$, and $\mathcal{P}$ denotes the set of parameters related to terrain properties.

B. Terramechanics Model

The terramechanics model is adopted from [16]. The model is briefly summarized here for completeness, and the interested reader is referred to [16] for a detailed explanation and validation.

In this model, the Soil Contact Model (SCM) is regarded as the ground truth. The SCM discretizes the interaction between tires and terrain and extends the Bekker model to simulate dynamic operations in 3D [24]. The SCM is commonly used to evaluate terramechanics with higher fidelity than the standard Bekker model and higher computational efficiency than the discrete element models [14]. However, its computation time is still too long and its lack of twice continuous differentiability is unsuitable for direct usage in MPC.

To create an optimal control-oriented terramechanics model, a shallow neural network is trained to emulate the behavior of the SCM. Latin hypercube sampling (LHS) is utilized to maximize the minimum distance between the inputs of the neural network model. To achieve satisfactory performance, 10,000 input samples are collected. The training data is obtained from the SCM terrain implementation using the high-fidelity simulation software Project Chrono [25]. The neural network model employs a hyperbolic tangent sigmoid transfer function to ensure twice continuous differentiability. The neural network is trained using the MATLAB deep learning toolbox. For the sake of computational efficiency and high fidelity, 2 hidden layers are chosen for the neural network, consisting of 14 neurons – 12 in the first layer and 2 in the second layer. For further details and validation, please see [16].

C. Terrain Estimator

The terrain estimator is adopted from [16]. Using the unscented Kalman filter (UKF), the estimator estimates the sinkage exponent n , which has been identified as the most sensitive parameter in the terramechanics model.

The UKF relies on a model for prior estimation that incorporates the vehicle dynamics and the neural network terramechanics model described in Sec II-A and II-B. In the UKF model, the state vector includes two sinkage exponents for the front and rear wheels in addition to the first six states specified in Eq. (1). The control inputs consist of the tire steering angle δ_f , longitudinal acceleration a_x , and the slip ratios α_f and α_r at the front and rear wheels, respectively. The UKF determines the most accurate estimation of the current vehicle states, including the sinkage exponents at the front and rear wheels, by comparing the model's prediction with the actual measurements of the first six states. Enhancing the estimation of sinkage exponents improves the prediction accuracy in MPC. Further analysis and discussion of the UKF implementation and simulation findings can be found in [26].

D. MPC Formulation

This section presents the formulation of the MPC framework, which is formulated as a single-level architecture, enabling simultaneous trajectory planning and tracking by directly mapping the control to the vehicle. However, only the trajectory planning aspect is utilized downstream in the

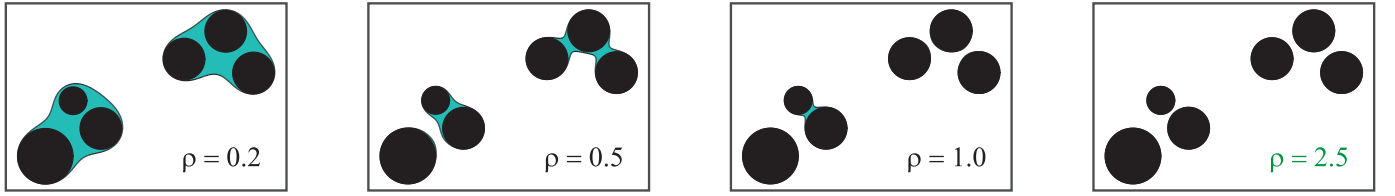


Fig. 2. Contour plots depict modified Kreisselmeier-Steinhaus function aggregated constraints: black indicates original obstacles, cyan represents the infeasible region generated by the aggregated constraints, and white denotes traversable areas. The ρ value increases from left to right, causing the infeasible area to converge towards the original obstacles.

larger framework, and the tracking is performed using a local controller to decouple the update rates of the trajectory planning and tracking and achieve a higher update rate for the latter as described in Sec. II-E.

The trajectories are obtained by solving the following optimal control problem (OCP):

$$\begin{aligned} \underset{\xi, \zeta}{\text{minimize}} \quad & J = \int_{t_0}^{t_f} \mathcal{I}[\xi(t), \zeta(t)] dt \end{aligned} \quad (10)$$

$$\text{subject to} \quad \dot{\xi}(t) = \mathcal{V}[\xi(t), \zeta(t)] \quad (11)$$

$$\xi_{\min} \leq \xi(t) \leq \xi_{\max} \quad (12)$$

$$\zeta_{\min} \leq \zeta(t) \leq \zeta_{\max} \quad (13)$$

$$\mathcal{G}[\xi] \leq \mathbf{0} \quad (14)$$

Here, J is the cost function, $\mathcal{I}$ is the stage cost of states and control inputs, $\mathcal{V}$ is the vehicle dynamics, and $\mathcal{G}$ is the vector of constraints for obstacle avoidance. The states and control trajectories are bounded by Eq. (12) and Eq. (13).

1) *State and Control Constraints*: Three state bounds are set as shown below.

$$x_{\min} \leq x \leq x_{\max} \quad (15)$$

$$u_{x_{\min}} \leq u_x \leq u_{x_{\max}} \quad (16)$$

$$a_{x_{\min}} \leq a_x \leq a_{x_{\max}} \quad (17)$$

The bounds of control inputs are set to be smaller than the actuator limits on board the vehicle.

$$\dot{\delta}_{f_{\min}} \leq \dot{\delta}_f \leq \dot{\delta}_{f_{\max}} \quad (18)$$

$$j_{x_{\min}} \leq j_x \leq j_{x_{\max}} \quad (19)$$

To generate a collision-free trajectory, hard constraints are imposed on the position of the vehicle as follows:

$$d_{\text{obs}}^i(t) = (x(t) - x_{\text{obs}}^i)^2 + (y(t) - y_{\text{obs}}^i)^2 - (r_{\text{obs}}^i + \epsilon)^2 \geq 0 \quad (20)$$

The i^{th} obstacle is located at the coordinates $(x_{\text{obs}}^i, y_{\text{obs}}^i)$, and its radius r_{obs}^i includes both the radius of the smallest circle encompassing the obstacle and the vehicle half-width. The safety margin ϵ accounts for tracking errors caused by the local controllers.

2) *Constraint Aggregation*: The number of hard constraints on obstacle avoidance, as indicated in Eq. (20), depends on the number of obstacles. In a clustered environment, especially if an occupancy grid is used with a fine resolution, a large number of hard constraints may arise, leading to computational

challenges. To alleviate this issue, a modified Kreisselmeier-Steinhaus function [27] is implemented to aggregate the multiple hard constraints in Eq. (14) into a single one.

$$\mathcal{G}(\xi) = \sum_{i=1}^{n_{\text{obs}}} \sum_{j=1}^{n_{\text{ds}}} e^{-\rho d_{\text{obs}}^i(\mathcal{T}(j))} - 1 \leq 0 \quad \forall \rho > 0 \quad (21)$$

The constraint is written by discretizing the optimal control problem without loss of generality. Here, n_{ds} represents the number of discrete states along the trajectory, while $\mathcal{T}$ denotes the mapping from the j^{th} point on the trajectory to time t in Eq. (20). n_{obs} is the number of obstacles, and ρ controls the conservativeness as the constraint aggregation curvature parameter.

To establish the conservativeness of this approach, one needs to establish that if the aggregated constraint is satisfied, then all individual constraints are satisfied. To prove that, let $g(i, j) = e^{-\rho d_{\text{obs}}^i(\mathcal{T}(j))}$. Note that $g(i, j) > 0$. Then, $\forall i \in [1, n_{\text{obs}}]$ and $\forall j \in [1, n_{\text{ds}}]$

$$\sum_{i=1}^{n_{\text{obs}}} \sum_{j=1}^{n_{\text{ds}}} g(i, j) > \sum_{j=1}^{n_{\text{ds}}} g(i, j) > g(i, j) \quad (22)$$

Assume that the aggregated constraint is satisfied; i.e., $\sum_{i=1}^{n_{\text{obs}}} \sum_{j=1}^{n_{\text{ds}}} e^{-\rho d_{\text{obs}}^i(\mathcal{T}(j))} \leq 1$. Then,

$$g(i, j) \leq 1 \quad \forall i \in [1, n_{\text{obs}}] \quad \forall j \in [1, n_{\text{ds}}] \quad (23)$$

Given $\rho > 0$, Eq. (20) implies $d_{\text{obs}}^i(\mathcal{T}(j)) \geq 0$. Thus, if the aggregated constraint is satisfied, all the individual constraints in Eq. (20) are satisfied simultaneously.

Fig. 2 illustrates four aggregated constraints. With increasing ρ , the cyan area (infeasible region) converges towards the black area (original obstacle region). Increasing ρ brings the aggregation closer to the exact hard constraints, whereas decreasing ρ results in a more conservative aggregation. For all $\rho > 0$, the aggregation is guaranteed to yield a more conservative infeasible area compared to the set of multiple constraints. Here, $\rho = 2.5$ is selected to align the infeasible region with the obstacle region. Reducing the number of hard constraints to one, regardless of the number of obstacles in the environment, can significantly decrease the OCP solve time.

3) *Cost Function*: Three components contribute to the cost function:

$$J_{\text{cs}} = \int_{t_0}^{t_f} (w_{\dot{\delta}_f} \dot{\delta}_f(t)^2 + w_{j_x} j_x(t)^2 + w_y y(t)^2) dt \quad (24)$$

$$J_{\text{obs}}^i = \int_{t_0}^{t_f} (\tanh(a_t k_{\text{obs}}(t)) + 1) dt \quad (25)$$

$$k_{\text{obs}}(t) = \left(\frac{x(t) - x_{\text{obs}}^i}{r_{\text{obs}}^i + \epsilon} \right)^2 + \left(\frac{y(t) - y_{\text{obs}}^i}{r_{\text{obs}}^i + \epsilon} \right)^2 \quad (26)$$

$$J_{\text{goal}} = \frac{(x(t_f) - x_f)^2 + (y(t_f) - y_f)^2}{(x(t_0) - x_f)^2 + (y(t_0) - y_f)^2} \quad (27)$$

Eq. (24) is the cost for the controls and states. The first two terms regularize the control effort. The third term is optional and keeps the vehicle as closely aligned with the centerline as possible if such behavior is desirable, where the centerline is the line connecting the start and goal points. Without loss of generality, the centerline is assumed to be aligned with the y-axis in Eq. (24) and the examples considered in this paper. Eq. (25) are the soft constraints for obstacle avoidance. They encourage, but not guarantee, obstacle avoidance. The parameter a_t controls the size of the area that the cost function penalizes and is set to $a_t = -1.3$ in this experiment. These constraints are designed to significantly increase the cost as the vehicle approaches the obstacle borders while having negligible impact when the vehicle is far from them. The surface plot in **Evaluate Costs** in Fig. 1 shows the dramatic increase in cost when the vehicle is near or inside the boundaries. Eq. (27) minimizes the distance between the terminal position and the goal position with respect to the distance between the initial position and the goal position. Normalization is used to avoid ill-conditioned problems when the vehicle is far away from the goal point. This cost encourages the vehicle to reach the goal position as quickly as possible.

By combining the three costs, the OCP cost in Eq. (10) is written as follows:

$$J = w_1 J_{\text{cs}} + w_2 \sum_{i=0}^{n_{\text{obs}}} J_{\text{obs}}^i + w_3 J_{\text{goal}} \quad (28)$$

with ω_j , $j = 1, 2, 3$, being the weights of the respective terms. The determination of the weights is a design choice based on how important the cost components are relative to each other. Here, the weights are chosen so that each term has the same order of magnitude and they are tuned to achieve satisfactory performance.

4) *OCP Settings*: The prediction horizon t_f is set to 3 s. In contrast to [16], a fixed horizon is used, because it proves to be more versatile. Notably, no additional tuning is necessary when the scenario changes. The OCP is transcribed into a non-linear programming using NLOptControl [28], [29] and solved through Ipopt [23]. The backward Euler integration scheme is used for the transcription with 10 points. Unlike [16], the Legendre-Gauss-Radau approach is discouraged due to potential fatal failures resulting from uneven time step discretization, particularly with larger time steps in long-distance scenarios. Conversely, backward Euler avoids this issue due to its uniform discretization. Although this integration scheme may introduce larger errors compared to higher order integration schemes [30], these errors can be mitigated through frequent updates. Moreover, the backward Euler integration scheme can offer a higher chance of converging to the optimal solution for short periods of allowed computational time [31]. With this solution strategy, the algorithm can satisfy the desired update rate of 10 Hz on the target platform (see Sec. II-F),

generating a new solution every 0.1 s after gathering the most recent information on vehicle states.

E. Local Controller

The vehicle follows the trajectories planned by MPC using two proportional integral derivative (PID) controllers; one for the path and the other for the speed.

The PID controller for path tracking is expressed as follows:

$$\delta_{\text{PID}}(t) = K_{p,l} e_l(t) + K_{i,l} \int e_l(t) dt + K_{d,l} \frac{de_l(t)}{dt} \quad (29)$$

where δ_{PID} is the output of the PID controller with the gains $K_{p,l}$, $K_{i,l}$, and $K_{d,l}$. e_l is the path tracking error defined as the minimum distance between the planned path and the look-ahead point shown in Fig. 1-D. In Fig. 1-D, the look-ahead distance d_l is adaptive to the current speed by the following formula:

$$d_l(t) = \begin{cases} d_{l,p}(t), & \text{if } d_{l,\min} < d_{l,p}(t) \\ d_{l,\min}, & \text{otherwise} \end{cases} \quad (30)$$

$$d_{l,p}(t) = u(t)t_l \quad (31)$$

where $u(t)$ represents the current speed of the vehicle, and t_l denotes the look-ahead time. The look-ahead distance is lower bounded by $d_{l,\min}$ to prevent unstable behavior.

The steering angle command δ_f applied to the vehicle is obtained by saturating the output of the PID controller to obey the steering limits as follows:

$$\delta_f(t) = \begin{cases} \delta_{\text{PID}}(t), & \text{if } |\delta_{\text{PID}}(t)| < \delta_{f,\max} \\ \delta_{f,\max} \text{sign}(\delta_{\text{PID}}(t)), & \text{otherwise} \end{cases} \quad (32)$$

where $\delta_{f,\max}$ is the maximum magnitude of the steering angle.

The speed controller follows a similar logic. The PID controller is given by:

$$p_{\text{PID}}(t) = K_{p,u} e_u(t) + K_{i,u} \int e_u(t) dt + K_{d,u} \frac{de_u(t)}{dt} \quad (33)$$

where p_{PID} is the output of the speed tracker with the gains $K_{p,u}$, $K_{i,u}$, and $K_{d,u}$. The error in the speed tracker denoted as e_u , is defined as the difference between the current speed and the planned speed at the queried point. The PID output is saturated as follows:

$$p_u(t) = \begin{cases} p_{\text{PID}}(t), & \text{if } p_{\min} < p_{\text{PID}}(t) < p_{\max} \\ p_{\min}, & \text{if } p_{\text{PID}}(t) < p_{\min} \\ p_{\max}, & \text{otherwise} \end{cases} \quad (34)$$

p_u represents the percentage of throttle pedal position commanded to the vehicle. The percentage of throttle pedal position linearly ranges from 0 to 100%. A value of 0% means releasing the pedal and 100% represents full throttle. $p_{\min} = 0\%$ and $p_{\max} = 85\%$ are the minimum and maximum allowed throttle pedal positions.

The local controller operates at approximately 1 kHz.

F. Plant Vehicle

The testing vehicle utilized in this study is a gasoline-powered MRZR, which has a seating capacity of four passengers as depicted in Fig. 1-E. All experimental tests are performed with the vehicle operating in rear-wheel-driven mode. The MRZR is equipped with four motor controllers responsible for electronic control of the brake and throttle pedals, steering wheel, and gear. Two TORC motor controllers are responsible for throttle and brake actuation. An Allied Motion POW-R STEER Electric Power Assisted Steering (EPAS) system controls the steering actuation. An Elmo Twitter servo drive is in charge of the gear shifting. These four motor controllers communicate with each other via the CAN network.

An RR-N-140 navigation system measures the precise absolute and relative 3D (6 DoF) localization information of the vehicle. The navigation system consists of IMU and GPS positioning units, both installed on the cargo bed. Two antennas for the GPS unit are positioned on the front and rear sides of the vehicle. The vehicle has four wheel encoders mounted, one for each wheel. To enable remote emergency stop functionality, a radio receiver is placed on the cargo bed. The vehicle features three additional manual emergency stop buttons, which are situated in the dash panel box, the driver's side of the rear cargo bed, and the passenger-side front fender.

To safeguard the vehicular computer system, a customized Pelican case with an aluminum mounting plate and heatsink is fitted in the cargo bed. The on-board computer is a Neousys Nuvo-5108VTC with an Intel Core i7-6700TE processor. This computing system serves as the central hub for communication with the sensors, controllers, and an external computer used for optimization and control command generation. An external computer is used to run the presented algorithms. It is a Dell Precision 5500 Workstation laptop using Intel Core i7-10850H processor clocked at 2.7 GHz and 62.6 GB memory. The connection between the external computer and the computing system of the vehicle is established over ROS using Ethernet.

G. Planning and Control Framework

The summary of the overall planning and control framework is shown in Fig. 1. By feeding current vehicle states, obstacle data, and updated sinkage exponent to the MPC planner, the trajectory is optimized and sent to the local controller. Based on the optimized trajectory, the local controllers compute the appropriate steering and throttle commands and send them to the electronic controllers through the CAN Bus. Subsequently, the embedded EPAS and TORC motor controllers execute these commands to facilitate the vehicle's movement along the planned trajectory.

III. MPC ROBUSTNESS TESTING

To analyze the ability of the proposed MPC formulation, a robustness test is conducted by navigating a simulated vehicle through obstacle fields on deformable terrains. This test follows the same procedure as in [32] to allow comparisons to prior work. Specifically, the test is simulated using the Mississippi State University Autonomous Vehicle Simulator

TABLE I
SETTINGS FOR THE COMBINED SCENARIO

Description	Mean	Standard Deviation
Vehicle weight	1071.85 kg	77.66 kg
Initial X-Position in slalom	-31.75 m	0.34 m
Initial Y-Position in slalom	0 m	0.22 m
Goal X-Position in slalom	52 m	0.22 m
Goal Y-Position in slalom	0 m	0.22 m
Soil strength	85 psi	4.7 psi
Rain density	3 mm/hr	0.74 mm/hr
X-Position of roadblock	137 m	2.3 m
Y-Position of roadblock	0 m	0.23 m
Roadblock scale factor	1.25 m	0.17 m
Roadblock orientation	89.77°	10.33°
Obstacle density in random	2489 N/(km) ²	126 N/(km) ²

(MAVS), a tool for simulating vehicles with physics-based sensors in an off-road environment [33]. The complex scenario designed in [32] is considered, which features an obstacle field consisting of three regions with randomized elements as summarized in Table I. These elements are not only limited to the number of obstacles and their sizes, positions, and orientations, but also include vehicle weight, soil strength, and rain density, and follow Gaussian distributions. One sample realization of the obstacle field is plotted in Fig. 3-A. The first region forces the vehicle to perform a slalom. The second region has a single roadblock in the middle, allowing for a long straight run with single obstacle avoidance. The third region features a random obstacle field. The parameter "soil strength" in Table I governs the relationship between the terrain's material failure and effective stress. A higher value of soil strength indicates a reduced likelihood of shearing and slipping. The parameter "rain density" determines the intensity of rainfall, which limits the detection range of the vehicle's sensor for obstacles. A higher value of rain density indicates a reduced detection range. The vehicle's desired speed is a constant 9 m/s in all three regions, but can change if the local controller deems it necessary to better follow the reference path. The desired speed is consistent with the US Defense Advanced Research Projects Agency (DARPA)'s Phase 1 targets for high speed off-road autonomy in the RACER program, which are 5 m/s for complex terrains and 10 m/s for open fields. Here, the built-in local controller of MAVS is used. For further details of the scenario, please see [32].

In the robustness test, seven algorithms are used to navigate the vehicle through a complex scenario, enabling comparison with state-of-the-art approaches. As shown in Table II, the MPC formulations differ by 1) integration scheme, 2) use of hard and soft constraints, 3) use of KS aggregation, 4) fixed time horizon, and 5) fixed terminal constraints. MPC #1 adheres to the proposed MPC design outlined in Sec. II-D and simultaneously incorporates both hard and soft constraints for all obstacles. MPC #2 to MPC #6 are derived from MPC #1 for the ablation study, while MPC #7 represents the state-of-the-art algorithm from [16]. The last state-of-the-art benchmark utilizes the artificial potential field (APF) approach, which is demonstrated as one of the most robust algorithms in [32]. The APF design follows [32].

Fig. 3-B presents a bar chart depicting the solving time statistics for the various MPC formulations. MPC #1 exhibits

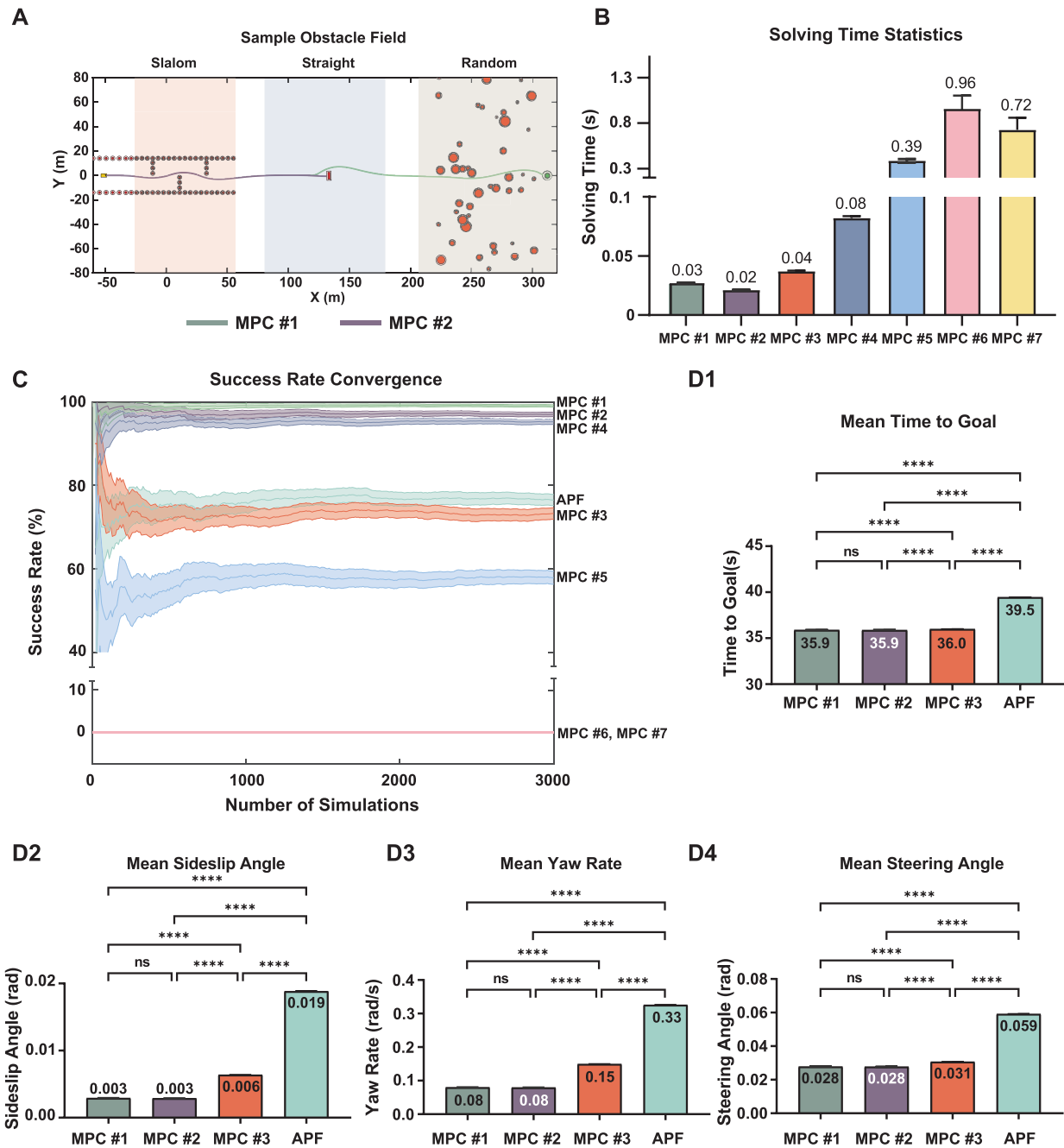


Fig. 3. The results of MPC robustness testing. (A) One example of a complex scenario with two paths planned by different algorithms. (B) Bar chart of the distributions of solving times. (C) Evolution of algorithms' success rates with increasing number of simulations. (D) Bar charts comparing the performance metrics. (D1) Statistical comparison of mean time to goal. (D2) Statistical comparison of mean sideslip angle). (D3) Statistical comparison of mean yaw rate). (D4) Statistical comparison of mean steering angle).

an average solving time of 27.2 ms, a slight increase compared to 21.3 ms for MPC #2. The inclusion of additional aggregated hard constraints marginally increases the solving time. MPC #3 has a solving time of 42.3 ms, increasing the solving time for MPC #1, as the gradients of the proposed soft constraints facilitate solution convergence. MPC #4 is designed to separately handle the hard constraints for each obstacle. Despite the manageable number of obstacles in the current scenario, the solving time increases further to 82 ms, approximately three times that of the proposed method. Aggregating the constraints significantly reduces computational load by

consolidating obstacle avoidance constraints, aiding the newly proposed MPC formulation in meeting real-time requirements. MPC #5 inevitably faces the challenge of scaling the problem because the travel time between the starting and intermediate goal points is substantial, necessitating a larger number of collocation points to ensure prediction accuracy. This results in a dramatic increase in solving time to 386 ms in this scenario. MPC #6 switches from the Backward Euler integration scheme to the LGR collocation method. It fails to perform effectively in off-road settings with an increased number of obstacles. The additional computational burden of collocation equality

TABLE II
SUMMARY OF THE SETTINGS FROM MPC #1 TO MPC #7

MPC #	Integration Scheme	Constraints	KS	Fixed Horizon	Terminal Constraints
1	BkwEuler	Both	✓	✓	×
2	BkwEuler	Soft	✓	✓	×
3	BkwEuler	Hard	✓	✓	×
4	BkwEuler	Both	×	✓	×
5	BkwEuler	Both	✓	×	✓
6	LGR	Both	✓	✓	×
7 [16]	LGR	Hard	×	×	✓

constraints increases the solution time to 957 ms. The method described in [16], denoted as MPC #7, also fails to achieve real-time performance. The average solving time for this approach is 725 ms.

Fig. 3-C shows the success rates for each algorithm, with 3,000 simulations conducted to ensure the metric's convergence. Success is defined as avoiding collisions and reaching the goal within a 4-meter tolerance. The upper and lower bounds represent the 95% confidence interval for the success rate. The proposed method, MPC #1, achieves a success rate of 99.03%, whereas MPC #2 exhibits a success rate of 97.00%. Fig. 3-A illustrates a typical case where MPC #1 succeeds while MPC #2 does not. When the vehicle enters the Straight portion and encounters a single obstacle detected by its sensors, MPC #2 plans a trajectory that directly intersects the obstacle, as it cannot differentiate between a suboptimal trajectory and an infeasible one. In contrast, MPC #1, incorporating hard constraints, successfully avoids obstacles and completes the task. However, MPC becomes more vulnerable when using only hard constraints, as evidenced by the 73.23% success rate of MPC #3. This is because the planner frequently designs trajectories close to the obstacles' boundaries, where even minor tracking errors can lead to failure. MPC #4 differs from MPC #1 only in the constraint aggregation, leading to lower success rate of 95.2%. The decrease in the success rate is because the aggregated constraint is more conservative than the original constraints, resulting in safer trajectories. Compared to MPC #1, MPC #5 includes terminal constraints, planning from the start position to the end. Consequently, the time interval between two collocation points becomes inevitably large at the beginning. This results in the "corner cutting" effect, where waypoints are collision-free, but the connecting line segments are not, potentially causing the vehicle to unintentionally hit obstacles. This issue reduces the success rate to 57.87%. To mitigate this issue, MPC #7 from [16] utilizes LGR collocation points. These points densify the time horizon around the start and end points, effectively eliminating the "corner cutting" effect as the vehicle starts to perform obstacle avoidance maneuvers. However, this adjustment makes the problem too complex to solve, preventing the planner from generating viable trajectories. Consequently, the success rate drops drastically to 0% in complex scenarios. MPC #6 inherits the same issue as MPC #7, where the planner fails to predict feasible trajectories, resulting also in a success rate of 0%. The Artificial Potential Field (APF) approach achieves a higher success rate of 76.47% compared to MPC #4. However, this

is still significantly lower than the 99.03% success rate of the proposed MPC #1.

Four metrics are introduced to compare the vehicle performance in the successful runs, which are also used in later discussion in Sec. IV. The four metrics and the rationale behind them are summarized as follows:

- 1) **Time to goal:** This indicator represents the duration of the vehicle's travel time to reach the destination. A smaller value denotes higher vehicle mobility and is thus desirable.
- 2) **Mean absolute sideslip angle:** This measure is one of the main indicators for the yaw stability of a vehicle [34]. A lower sideslip angle indicates a more stable behavior and is thus desirable.
- 3) **Mean absolute yaw rate:** This metric is another main indicator for the yaw stability of a vehicle [34]. A smaller value is desirable, as it represents a less aggressive and more stable maneuver.
- 4) **Mean absolute steering angle:** This metric represents the control effort. A smaller value indicates that the algorithm spends less effort to navigate the vehicle and is desirable.

To evaluate the four metrics, only those scenarios where all algorithms succeed are selected. Because MPC #5, MPC #6, and MPC #7 have significantly lower success rates than the other candidates, they are excluded from this study. While MPC #4 performs similarly to MPC #1 in terms of navigation to goal points, their primary difference lies in solving time. Therefore, MPC #4 is also excluded from this study. Thus, the study focuses on the performance of four algorithms: MPC #1, MPC #2, MPC #3, and the Artificial Potential Field (APF) method. The comparisons are conducted using Analysis of Variance (ANOVA), with results displayed in Fig. 3-D. In Fig. 3, the asterisks over the plots demonstrate the statistical significance level between the planners, namely, * indicates $p < 0.05$, ** indicates $p < 0.01$, *** indicates $p < 0.001$, and **** indicates $p < 0.0001$. Across all metrics, there are no statistically significant differences between MPC #1 and MPC #2. This suggests that the obstacle avoidance maneuvers are primarily influenced by the soft constraints.

Hence, MPC #1 outperforms MPC #2 only in safety, with a 99.03% vs 97.00% success rate. The two algorithms exhibit statistical differences compared to MPC #3 in the metrics of mean time to goal, mean absolute steering angle, mean absolute sideslip angle, and mean absolute yaw rate. These differences arise from sudden changes in optimal trajectories when the vehicle experiences significant tracking errors due to the stochastic settings of the robustness test. MPC #3 needs to design aggressive turns to prevent collisions with obstacles. In contrast, MPC #2 and MPC #1 allocate extra space to the obstacles based on the design of the cost function and take less risk. Therefore, even if the tracking errors become significant, the two algorithms do not require aggressive correction maneuvers.

In all metrics, MPC #1, 2, and 3 provide more optimal solutions compared to the APF solutions. This is evidenced by the superior performance in mean absolute yaw rate and mean absolute sideslip angle comparisons, indicating that

TABLE III
SETTINGS FOR THE THREE OBSTACLE AVOIDANCE SCENARIO

Description	Value
Obstacle I position	[52.718, -1] m
Obstacle II position	[72.718, 1] m
Obstacle III position	[92.718, -2] m
Obstacle radius	2.5 m
Obstacle safety margin	0.75 m
Vehicle starting position	[2.718, 0] m
Vehicle goal position	[112.718, 0] m
Maximum allowable longitudinal speed	10 m/s
Radius of goal tolerance	7.2 m
Soil deformation modulus	1515000 (N/m ⁿ⁺¹)
Shear deformation modulus	0.025
Cohesion	1700 (Pa)
Friction angle	0.5061 (rad)

A



B

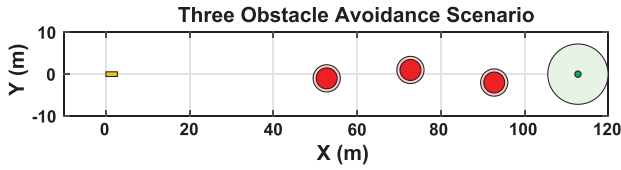


Fig. 4. Overview of the test scenario: (A) An aerial view of the test site. The approximate relative location of the three virtually placed obstacles are shown as red disks, and the safety margin of each is shown as a transparent red disk. The green dot and the ring around it are the goal position and tolerance, respectively. (B) The diagram for the three obstacle avoidance scenario. The yellow rectangle represents the vehicle at the initial position. The red disks are the obstacles with the safety margin shown as the outer rings. The green disk is the goal point, and the outer ring of the goal is the tolerance range.

the vehicle executes less aggressive maneuvers, resulting in smoother turns. The penalty on control actions in MPC further results in reduced steering effort throughout the simulations. Additionally, the MPC algorithms exhibit significantly shorter time-to-goal metrics compared to APF, demonstrating that the MPC solutions provide greater vehicle mobility.

In conclusion, incorporating terminal cost and employing the backward Euler collocation method significantly enhance the ability to find viable solutions in complex environments. The use of constraint aggregation can substantially improve computational performance. Combining hard and soft constraints leads to less aggressive maneuvers and a higher success rate. These features are crucial for achieving real-time operation in complex off-road environments.

IV. EXPERIMENTAL RESULTS

The physical experiments were performed at Bundy Hill Off Road Recreational Park in Jerome, MI. Fig. 4-A shows an

aerial view of the vehicle at the test location with virtual obstacles. The obstacles are placed relative to the initial position and heading of the vehicle in each run, thereby ensuring their consistent placement. Fig. 4-B presents a diagram providing a clearer top view of the scenario. The detailed settings are summarized in Table III.

Three MPC formulations are considered in the experiments:

- 1) **Pacejka MPC**: This MPC formulation uses the Pacejka tire model explained in Sec. II-A. Thus, it assumes a rigid terrain and serves as a benchmark.
- 2) **Terrain-Aware MPC**: This MPC uses the proposed formulation without the terrain estimator.
- 3) **Terrain-Adaptive MPC**: This MPC uses the proposed formulation with the terrain estimator to continuously update the sinkage exponent in real time.

The constraint aggregation discussed in Sec. II-D is not considered in the experiments due to the small number of obstacles and their significant separation in distance from each other.

Three experiments were performed on three different days and are presented in three sections below. The first experiment compares Pacejka MPC and Terrain-Aware MPC, whereas the second experiment evaluates the impact of adding the terrain estimator in Terrain-Adaptive MPC. The last experiment studies the effect of real-time MPC formulation.

A. Terrain-Aware MPC

This section evaluates the Terrain-Aware MPC vis-a-vis the Pacejka MPC. All tests are done on the same day to ensure comparable weather and terrain conditions.

The testing site is characterized as sandy loam terrain, with the corresponding terrain parameters provided in Table III, sourced from [35]. The soil sinkage exponent is set to be 0.8 which is greater than the original value in [35] to account for the softer terrain caused by moisture content due to the substantial rainfall preceding the testing date.

Ten tests are conducted per formulation to demonstrate the disparities between the two formulations despite the inherent uncertainties in the test. The resulting paths of Terrain-Aware MPC and Pacejka MPC are plotted in Fig. 5-A1 and A2. Notably, while the Terrain-Aware MPC achieves success in every run, the Pacejka MPC experiences a failure in one of the trials, losing control and veering across the obstacle. This failure case is shown in red in Fig. 5-A2. Also, paths generated by Terrain-Aware MPC are more consistent than those produced by Pacejka MPC, showing that Terrain-Aware MPC has better control over the vehicle on deformable terrain.

A detailed statistical analysis is summarized in Fig. 5-B1 ~ B6. As seen in Fig. 5-A1 and A2, there is a linear acceleration zone at the beginning, where the behavior of both algorithms is similar. This acceleration region is excluded from the statistical analysis. Specifically, the metrics are calculated after the vehicle reaches the point at $x = 20$ m. The black circles and diamonds are the actual measured values from the experiments. In addition to the metrics described in III, a new metric called “path variance” is analyzed, as well, to assess the consistency

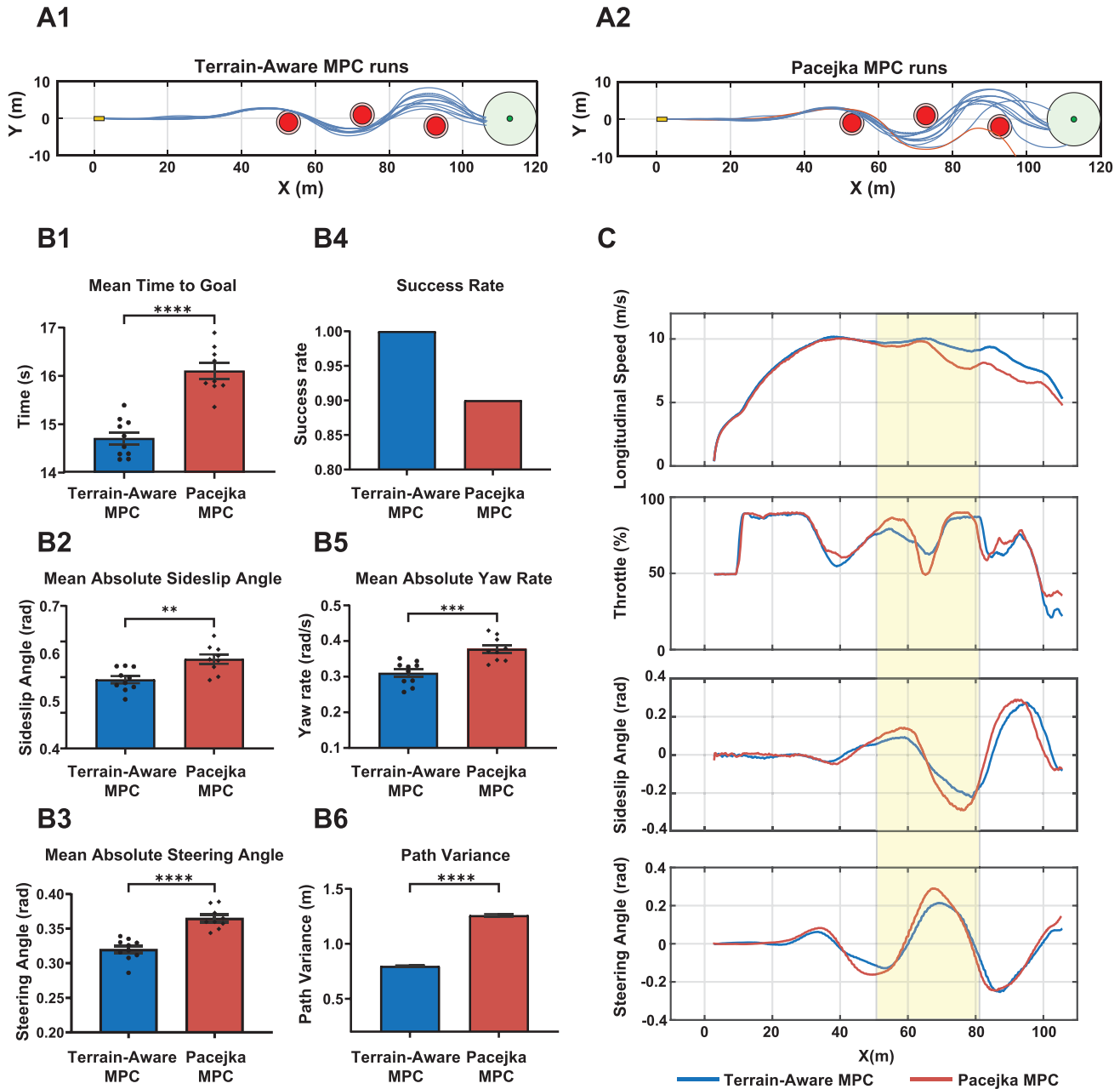


Fig. 5. Vehicle performance in Terrain-Aware formulation tests: (A1) Vehicle path in Terrain-Aware MPC runs. (A2) Vehicle path in Pacejka MPC runs. (B1-B6) Statistical comparisons between Pacejka and Terrain-Aware MPC for the ten runs. The bar chart values and the error bar values are the mean values and standard errors of the measured metrics. (B1) Statistical comparison in mean time to goal. (B2) Statistical comparison in mean absolute sideslip angle. (B3) Statistical comparison in mean absolute steering angle. (B4) Statistical comparison in success rate. (B5) Statistical comparison in mean absolute yaw rate. (B6) Statistical comparison in path variance. (C) Vehicle response under the two formulations. The four subplots utilize the same x-axis.

between runs. It demonstrates whether the vehicle's behavior is consistent between the runs. Less path variance indicates that the algorithm has better control over the vehicle and the outcomes are more predictable despite the uncontrollable uncertainties in the experiments.

Significant statistical differences are observed across all the metrics in the plot. These findings indicate that, on average, in the considered scenario, Terrain-Aware MPC produces trajectories with 23.4% less mean absolute sideslip angle, 17.8% less mean absolute yaw rate, and 23.8% less mean absolute steering angle, while achieving a 1.5 s faster time to reach the target. These results demonstrate Terrain-Aware MPC's

enhanced performance due to a better understanding of the mobility limits. In conclusion, when operating on deformable terrain, Terrain-Aware MPC outperforms Pacejka MPC in this scenario.

Fig. 5-C presents four plots: longitudinal speed, throttle position, sideslip angle, and steering angle, all versus the X position. These plots serve to illustrate why the terrain-aware formulation outperforms the benchmark on deformable terrain. The average values of all successful runs are utilized in creating these plots, allowing for an analysis of the algorithms' averaged behavior. The advantages of Terrain-Aware MPC become apparent in the yellow-shaded region, which

corresponds to the area when the vehicle encounters the first and second obstacles. Although the difference in longitudinal speed between the two algorithms is minor in this area, Terrain-Aware MPC shows a distinct reduction in throttle and steering angle. With a better understanding of the limitations imposed by lateral tire forces on the deformable ground, Terrain-Aware MPC chooses a less aggressive approach in that region. This advantage provides the vehicle with a larger operational range and increased stability before confronting the second and third obstacles. Conversely, Pacejka MPC overestimates the lateral tire forces, causing the vehicle to exceed its limits. Consequently, the vehicle spins and loses mobility when passing the second and third obstacles.

From the previous discussions, the Terrain-Aware formulation does, to some extent, yield benefits when operating the vehicle on deformable terrains, but drawbacks still exist:

- 1) The sinkage exponent value used is a naive approximation intended to account for the terrain's softness without any validation.
- 2) Upon the vehicle passing the third obstacle, the sideslip angle in Terrain-Aware MPC increases and is close to that of Pacejka MPC. Consequently, the vehicle continues to lose control at this stage. This observation is further substantiated by the path variance depicted in Fig. 5-A2. The paths demonstrate a significant level of consistency until reaching the third obstacle, beyond which they diverge before converging again near the goal.

These findings indicate that there is still potential for the algorithm to enhance its understanding of vehicle dynamics. To mitigate this gap, the terrain-adaptive formulation is tested in Sec. IV-B.

B. Terrain-Adaptive MPC

This section begins by contrasting three algorithms: Terrain-Adaptive MPC, Pacejka MPC, and Terrain-Aware MPC with a poor guess for the sinkage exponent, namely $n = 1.0$. Terrain-Adaptive MPC adds the sinkage exponent estimation component by using the UKF described in Sec. II-C and adopts the whole framework in Sec. II-G. The selections for the terrain parameters are the same as those in Sec. IV-A except for the sinkage exponent value, which was changed from 0.8 to 1.0 because of two reasons: 1) By intentionally setting the sinkage exponent to a wrong value we aimed to test whether a poorly parameterized terrain-aware formulation can still perform better than Pacejka MPC and thus whether an online estimation is needed at all. 2) This wrong value also serves as the initial guess in the terrain estimator. By using an unreasonably high initial guess far away from the correct value, we aim to stress test the convergence of the estimator.

The path plots are given in Fig. 6-A1 ~ A4. In this test, every algorithm successfully steers the vehicle to the desired goal location. The path plots indicate that Terrain-Adaptive MPC exhibits more consistent paths, whereas the other two algorithms show larger variations in paths.

The behavior of the terrain estimator is shown in Fig. 6-B, where the centerline represents the mean value of the estimation, and the upper and bottom lines represent one standard

deviation above and below the mean value, respectively. The nearly flat region of sinkage exponent between $x = 0$ m and $x = 40$ m is due to the lack of lateral dynamic excitation in the straight acceleration process. The sinkage exponent does not fully converge by the end of the runs, which can be explained by the short period of the experiment and the local variations of the surface. Nevertheless, the estimation moves rapidly to a lower value than the initial one as expected once the vehicle initiates its obstacle avoidance maneuvers.

To showcase the full benefit of terrain adaptation, Terrain-Aware MPC is run once more, this time utilizing the average estimated sinkage exponent from the terrain estimator in the final period of the experiment. Specifically, the final value is determined by averaging the sinkage exponent estimates in the orange zone in Fig. 6-B. Thus, a fourth formulation is created by using Terrain-Aware MPC with the sinkage exponent of $n = 0.56$. The path plots with this MPC formulation are shown in Fig. 6-B.

Then, a statistical analysis is carried out for the four groups in Fig. 7-A1 ~ A5. Due to the increased number of groups in this experiment compared to the previous one, ten repetitions of the test for each algorithm is insufficient to achieve a reasonable statistical power for ANOVA with all four groups. However, increasing the number of tests per algorithm was infeasible due to the need to finish all tests in a single day to avoid the day-to-day variations of the terrain conditions. Therefore, to reveal the statistically significant differences, the Newman-Keuls posthoc test is used to perform pairwise comparisons [36].

A summary of the examination of five metrics is provided below. Terrain-Aware ($n = 0.56$) and Terrain-Adaptive MPC are categorized as the proposed algorithms while the other two are classified as the benchmarks.

- 1) **Time to goal:** In comparison to the benchmarks, the proposed algorithms achieve greater mobility as characterized by a smaller time to goal. The average separation between these two groups is about 0.5 s. Given the same permitted top speed and the short track distance, an improvement of 0.5 s is significant.
- 2) **Mean absolute sideslip angle:** Terrain-Aware ($n = 0.56$) MPC has the lowest metric and shows statistically significant differences with Pacejka and Terrain-Aware ($n = 1.0$) MPC. Terrain-Adaptive MPC has the second smallest metric and shows a statistically significant difference with Pacejka MPC. According to these findings, the vehicle performs better in terms of stability and drifts less when operating with Terrain-Adaptive and Terrain-Aware ($n = 0.56$) MPC.
- 3) **Mean absolute yaw rate:** The proposed algorithms produce smaller turns than the benchmarks. The similar statistically significant differences are observed as in the mean absolute sideslip angle metric. Thus, with improved parameterization of the terramechanics model, the algorithm intentionally decreases the aggressiveness when turning.
- 4) **Mean absolute steering angle:** The vehicle controlled by the proposed algorithms makes smoother turns. From the statistical analysis, the proposed groups achieve smaller

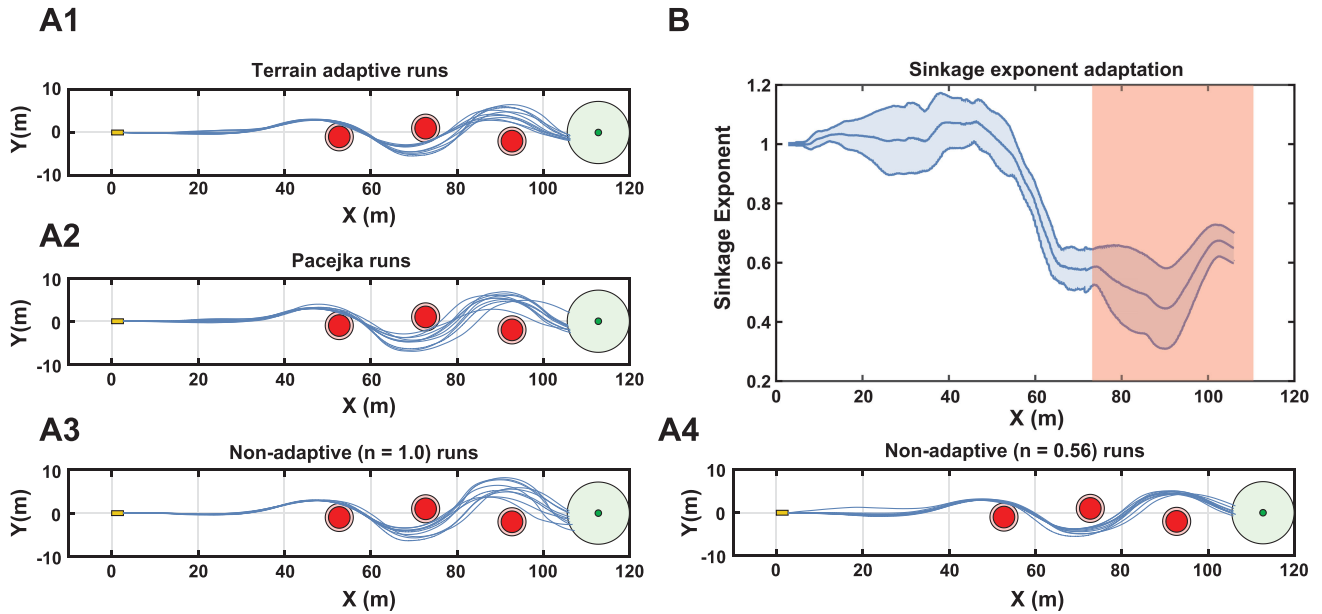


Fig. 6. (A1) Vehicle paths with Terrain-Adaptive MPC. (A2) Vehicle paths with Pacejka MPC. (A3) Vehicle paths with Terrain-Aware MPC ($n = 1.0$). (A4) Vehicle paths with Terrain-Aware MPC ($n = 0.56$). (B) Sinkage exponent (n) estimation and vehicle path with Terrain-Aware MPC ($n = 0.56$).

mean absolute steering angles and exhibit statistically significant differences with the benchmarks.

- 5) **Mean path variance:** There are statistically significant differences between each group. The path variance of Terrain-Aware ($n = 0.56$) MPC is much lower than that of the other groups. Terrain-Adaptive MPC performs more consistently near the end of the track with the aid of the terrain estimator, outperforming the two benchmarks. The performance of this metric is comparable between Pacejka MPC and the Terrain-Aware ($n = 1.0$) MPC. With Pacejka MPC as the benchmark, they barely perform differently from one another by 4.7%, compared to a 27.0% difference for Terrain-Adaptive MPC and a 55.8% difference for Terrain-Aware ($n = 0.56$) MPC.

In Fig. 7-B, the yellow area is the key part where the controllers exhibit different behaviors, resulting in good or bad performance. Under Terrain-Aware ($n = 0.56$) MPC, a rapid decrease in the vehicle's engine power is observed at $x = 65$ m. Additionally, the steering angle of Terrain-Aware ($n = 0.56$) MPC is less aggressive than that of the other formulations. The vehicle reduces its speed and makes smoother turns, such that the maneuvers between the second and third obstacles require less lateral tire forces. Consequently, the sideslip angle decreases relative to the other formulations.

Two trajectories designed by Terrain-Aware ($n = 0.56$) MPC and Pacejka MPC are chosen to further explore the differences between the controllers. Because there are ten runs for each MPC, the chosen trajectories are the ones closest to the average behavior in terms of the path difference. The initial conditions of each trajectory are as similar as possible to allow for a fair comparison and the initial X position is near 65 m, which is the key position identified above. Fig. 7-C shows the trajectory differences between Pacejka MPC and Terrain-Aware ($n = 0.56$) MPC. The top figure

shows the path difference, while the bottom figure shows the longitudinal speed difference in the designed trajectories. Terrain-Aware ($n = 0.56$) MPC designs a path with less curvature. It intentionally drops the speed from 10 m/s to around 8.5 m/s, as the planner believes that the vehicle does not have enough friction force under the current speed to help the vehicle steer. The algorithm purposefully reduces the speed to prevent the vehicle from experiencing severe side slips. This explains why Terrain-Aware ($n = 0.56$) MPC has a smaller steering angle and throttle position. On the other hand, assuming a rigid terrain, Pacejka MPC plans a more aggressive turn and a slow-down by only 0.2 m/s.

Based on the aforementioned findings, two important conclusions can be drawn:

- 1) The performance of Terrain-Aware MPC can deteriorate and become even worse than Pacejka MPC if the terrain parameter is chosen incorrectly. Thus, terrain estimation becomes critical.
- 2) Even if the original guess is significantly wrong, the terrain estimator can correct the sinkage exponent sufficiently fast to make a difference in the short maneuvers considered. Thus, Terrain-Adaptive MPC outperforms the benchmarks through continuous correction of the sinkage exponent. Additionally, with a corrected sinkage exponent estimate, Terrain-Aware MPC can provide consistent behavior, demonstrating higher repeatability than the benchmarks despite many uncontrollable uncertainties in the experiment.

C. Real-Time Solving Challenge

This section extends the previous two experiments to demonstrate the scalability of the real-time capability of the proposed MPC formulation. In this experiment, three algorithms are studied, each running three times. The first

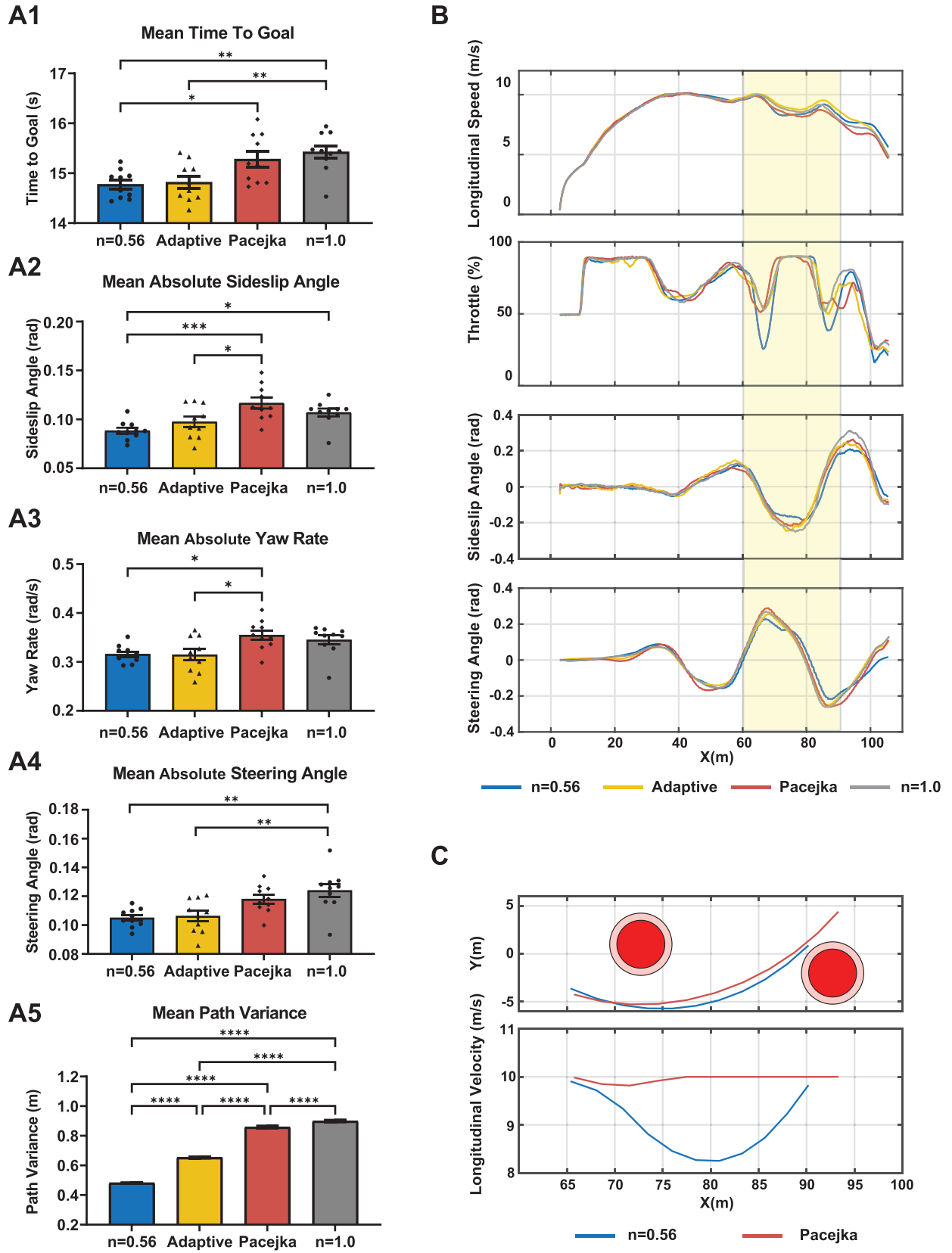


Fig. 7. Performance metrics and vehicle responses for the second experiment. (A1-A5) Statistical comparisons between Terrain-Aware ($n = 0.56$), Terrain-Adaptive, Pacejka and Terrain-Aware ($n = 1.0$) MPC for ten runs. (A1) Statistical comparisons in the mean time to goal. (A2) Statistical comparisons in the mean absolute sideslip angle. (A3) Statistical comparisons in the mean absolute yaw rate. (A4) Statistical comparisons in the mean absolute steering angle. (A5) Statistical comparisons in the mean path variance. (B) Vehicle responses under the four MPC formulations. (C) Planned trajectories by Terrain-Aware ($n = 0.56$) MPC and Pacejka MPC.

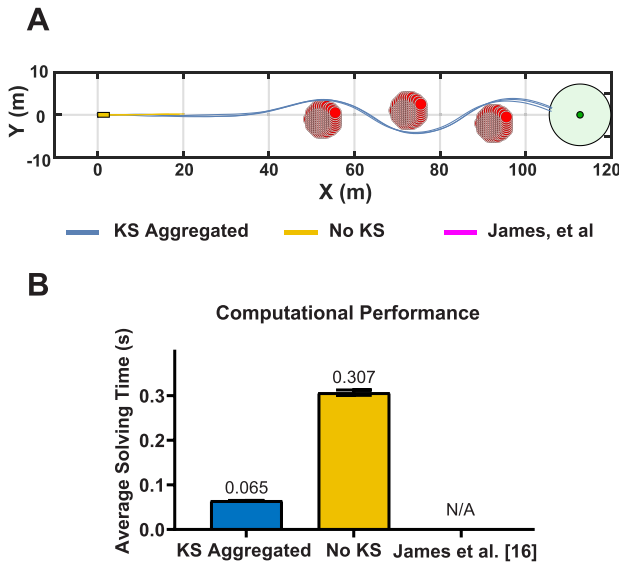


Fig. 8. Vehicle performance in real-time tests: (A) Vehicle path with and without KS aggregated and state-of-art formulations. (B) Statistical comparisons in average solving time. Only the optimal solutions' solving times are considered.

algorithm utilizes the fixed time horizon, backward Euler integration scheme, and KS aggregation. The second algorithm is similar to the first but does not use the KS aggregation. The last algorithm is the state-of-the-art described in [16]. The scenario remains the same as in the previous experiment except that the environmental complexity is increased. Specifically, each large obstacle (radius of 2.5 m) is discretized into 91 small obstacles with radius of 0.5 m.

The path plots are given in Fig. 8-A. In this test, the KS aggregated algorithm successfully steers the vehicle to the desired goal location in all three runs. Without the KS aggregation, the second algorithm needs to evaluate 273 nonlinear constraints for each collocation point, making it infeasible to optimize the trajectory in real time. As a result, the vehicle stops in front of the first cluster of obstacles. The path of the state-of-art does not appear in Fig. 8-A, as it cannot provide any solution even at the beginning.

The statistical analysis of solving time is summarized in Fig. 8-B. The solving time only reports the average time of the optimal solutions. Any infeasible solution's computational time is excluded. According to the bar chart, the average solving time of the KS-aggregated algorithm is around 65 ms, which satisfies the real-time requirement of a 10 Hz update rate. Without KS aggregation, the algorithm needs about 307 ms to generate a straight line, as shown in Fig. 8-A. Since the state-of-art algorithm cannot generate any optimal solution, the solving time is reported as N/A.

Thus, the unscalability of the state-of-art algorithm is demonstrated, as the algorithm cannot generate any optimal solution under the complex scenario. The No-KS aggregation formulation changes the integration scheme from LGR to backward Euler, makes the time horizon fixed, and adds the combined obstacle avoidance constraints. These changes allow the vehicle to start a straight motion at the beginning, but are not sufficient to complete the maneuver, as the algorithm fails

in front of the first obstacle cluster. By using the KS aggregation, the proposed algorithm successfully steers the vehicle to the goal, demonstrating the scalability of its performance.

V. CONCLUSION

In this work, a novel terrain-adaptive formulation is proposed for autonomous navigation of off-road vehicles on uncertain deformable terrains. The novelty of this formulation lies in its design for obstacle avoidance that combines hard and soft constraints, as well as its solution strategies. These novelties for the first time enable a more robust solution than the state of the art that can scale to cluttered environments while still maintaining an update rate of 10 Hz, a $5\times$ improvement over the state-of-the-art MPC formulation for deformable terrains. The robustness of the proposed formulation is evaluated in simulation against three benchmarks; namely, two state-of-the-art MPC formulations and an artificial potential field formulation. The results demonstrate that the suggested MPC formulation outperforms the benchmarks by achieving a higher success rate and performing smoother maneuvers while maintaining real-time computational speed. It is shown for the first time experimentally that considering the terramechanics in the MPC formulation significantly enhances the safety and effectiveness of autonomous navigation on a deformable terrain. A UKF-based terrain estimator is used in conjunction with the MPC to account for the key unknown terrain parameter, namely, sinkage exponent. By comparing the performance metrics of Terrain-Adaptive MPC to those of Pacejka MPC and Terrain-Aware MPC with an incorrect sinkage exponent, the importance and effectiveness of the terrain estimator in MPC-based navigation is validated experimentally for the first time. Experiments further indicate that employing the corrected sinkage exponent value in Terrain-Aware MPC significantly enhances performance. Finally, the algorithm is compared with state-of-art algorithms in a complex scenario, demonstrating significant improvements in the scalability of the proposed formulation. In light of these results, it is concluded that the proposed Terrain-Adaptive MPC framework is an important step toward increasing the mobility of autonomous vehicles on uncertain deformable terrains.

Expanding the algorithm to 3D terrains is identified as an important future direction. Further improving the computational speed is also critical to developing formulations that explicitly account for a wider range of uncertainties. The theoretical analysis of the algorithm's feasibility and robustness is another important direction for future research.

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REFERENCES

- [1] D. Osmankovic, A. Tahirovic, and G. Magnani, "All terrain vehicle path planning based on D* lite and MPC based planning paradigm in discrete space," in *Proc. IEEE Int. Conf. Adv. Intell. Mechatronics (AIM)*, Jul. 2017, pp. 334–339.

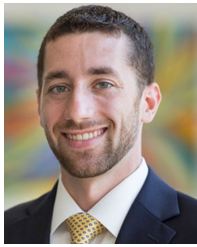
- [2] R. Song, Y. Liu, and R. Bucknall, "Smoothed A* algorithm for practical unmanned surface vehicle path planning," *Appl. Ocean Res.*, vol. 83, pp. 9–20, Feb. 2019.
- [3] S. A. Gunawan, G. N. P. Pratama, A. I. Cahyadi, B. Winduratna, Y. C. H. Yuwono, and O. Wahyunggoro, "Smoothed A-star algorithm for nonholonomic mobile robot path planning," in *Proc. Int. Conf. Inf. Commun. Technol. (ICOIAC)*, Jul. 2019, pp. 654–658.
- [4] C. Shen, S. Yu, and T. Ersal, "A three-phase framework for global path planning for nonholonomic autonomous vehicles on 3D terrains," *IFAC-PapersOnLine*, vol. 54, no. 20, pp. 160–165, 2021.
- [5] S. Dixit et al., "Trajectory planning for autonomous high-speed overtaking in structured environments using robust MPC," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 6, pp. 2310–2323, Jun. 2020.
- [6] S. Mata, A. Zubizarreta, and C. Pinto, "Robust tube-based model predictive control for lateral path tracking," *IEEE Trans. Intell. Veh.*, vol. 4, no. 4, pp. 569–577, Dec. 2019.
- [7] J. Wurts, J. L. Stein, and T. Ersal, "Collision imminent steering at high speed using nonlinear model predictive control," *IEEE Trans. Veh. Technol.*, vol. 69, no. 8, pp. 8278–8289, Aug. 2020.
- [8] J. Liu, P. Jayakumar, J. L. Stein, and T. Ersal, "Combined speed and steering control in high-speed autonomous ground vehicles for obstacle avoidance using model predictive control," *IEEE Trans. Veh. Technol.*, vol. 66, no. 10, pp. 8746–8763, Oct. 2017.
- [9] H. Febbo, J. Liu, P. Jayakumar, J. L. Stein, and T. Ersal, "Moving obstacle avoidance for large, high-speed autonomous ground vehicles," in *Proc. Amer. Control Conf. (ACC)*, May 2017, pp. 5568–5573.
- [10] S. Yu, C. Shen, and T. Ersal, "Nonlinear model predictive planning and control for high-speed autonomous vehicles on 3D terrains," *IFAC-PapersOnLine*, vol. 54, no. 20, pp. 412–417, 2021.
- [11] G. Ishigami, A. Miwa, K. Nagatani, and K. Yoshida, "Terramechanics-based model for steering maneuver of planetary exploration rovers on loose soil," *J. Field Robot.*, vol. 24, no. 3, pp. 233–250, Mar. 2007.
- [12] W. C. Smith, "Modeling of wheel-soil interaction for small ground vehicles operating on granular soil," Ph.D. dissertation, Dept. Mech. Eng., Univ. Michigan, Ann Arbor, MI, USA, 2014.
- [13] T. Guo, "Power consumption models for tracked and wheeled small unmanned ground vehicles on deformable terrains," Ph.D. dissertation, Dept. Mech. Eng., Univ. Michigan, Ann Arbor, MI, USA, 2016.
- [14] S. Taheri, C. Sandu, S. Taheri, E. Pinto, and D. Gorsich, "A technical survey on terramechanics models for tire-terrain interaction used in modeling and simulation of wheeled vehicles," *J. Terramech.*, vol. 57, pp. 1–22, Feb. 2015.
- [15] R. Krenn and G. Hirzinger, "SCM—A soil contact model for multi-body system simulations," in *Proc. Eur. Regional Conf. Int. Soc. Terrain-Vehicle Syst.*, 2009.
- [16] J. Dallas, M. P. Cole, P. Jayakumar, and T. Ersal, "Terrain adaptive trajectory planning and tracking on deformable terrains," *IEEE Trans. Veh. Technol.*, vol. 70, no. 11, pp. 11255–11268, Nov. 2021.
- [17] H. Taghavifar and S. Rakheja, "A novel terramechanics-based path-tracking control of terrain-based wheeled robot vehicle with matched-mismatched uncertainties," *IEEE Trans. Veh. Technol.*, vol. 69, no. 1, pp. 67–77, Jan. 2020.
- [18] H. Taghavifar, S. Rakheja, and G. Reina, "A novel optimal path-planning and following algorithm for wheeled robots on deformable terrains," *J. Terramech.*, vol. 96, pp. 147–157, Aug. 2021.
- [19] R. Zhang, W. Zhou, H. Liu, J. Gong, H. Chen, and A. Khajepour, "A terramechanics-based dynamic model for motion control of unmanned tracked vehicles," *IEEE Trans. Intell. Vehicles*, early access, May 28, 2024, doi: [10.1109/TIV.2024.3406582](https://doi.org/10.1109/TIV.2024.3406582).
- [20] P. Gupta, J. M. Smereka, and Y. Jia, "Reinforcement learning compensated model predictive control for off-road driving on unknown deformable terrain," 2024, *arXiv:2408.09253*.
- [21] J. Liu, P. Jayakumar, J. L. Stein, and T. Ersal, "A study on model fidelity for model predictive control-based obstacle avoidance in high-speed autonomous ground vehicles," *Vehicle Syst. Dyn.*, vol. 54, no. 11, pp. 1629–1650, Nov. 2016.
- [22] H. Pacejka, *Tire and Vehicle Dynamics*. Amsterdam, The Netherlands: Elsevier, 2005.
- [23] A. Wächter and L. T. Biegler, "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming," *Math. Program.*, vol. 106, no. 1, pp. 25–57, 2006.
- [24] R. Krenn and A. Gibbesch, "Soft soil contact modeling technique for multi-body system simulation," in *Trends in Computational Contact Mechanics*. Berlin, Germany: Springer, 2011, pp. 135–155.
- [25] A. Tasora et al., "Chrono: An open source multi-physics dynamics engine," in *Proc. Int. Conf. High Perform. Comput. Sci. Eng.*, Berlin, Germany: Springer, 2015, pp. 19–49.
- [26] J. Dallas et al., "Online terrain estimation for autonomous vehicles on deformable terrains," *J. Terramechanics*, vol. 91, pp. 11–22, Oct. 2020.
- [27] G. Kreisselmeier and R. Steinhauser, "Systematic control design by optimizing a vector performance index," in *Computer Aided Design of Control Systems*. Amsterdam, The Netherlands: Elsevier, 1980, pp. 113–117.
- [28] H. Febbo. (2022). *NLOptControl.jl*. Accessed: Feb. 6, 2022. [Online]. Available: <https://github.com/JuliaMPC/NLOptControl.jl>
- [29] H. Febbo, P. Jayakumar, J. L. Stein, and T. Ersal, "NLOptControl: A modeling language for solving optimal control problems," 2020, *arXiv:2003.00142*.
- [30] J. Wurts, J. L. Stein, and T. Ersal, "Design for real-time nonlinear model predictive control with application to collision imminent steering," *IEEE Trans. Control Syst. Technol.*, vol. 30, no. 6, pp. 2450–2465, Nov. 2022.
- [31] H. Febbo, "Real-time trajectory planning to enable safe and performant automated vehicles operating in unknown dynamic environments," Ph.D. dissertation, Dept. Mech. Eng., Univ. Michigan, Ann Arbor, MI, USA, 2023.
- [32] M. Cole, K. B. Kulkarni, J. Ewing, S. Tau, C. Goodin, and P. Jayakumar, "Exploration of robust and intelligent navigation algorithms to ensure off-road autonomous vehicle mobility," *Int. J. Vehicle Perform.*, vol. 10, no. 3, pp. 239–267, 2024.
- [33] C. Hudson, C. Goodin, Z. Miller, W. Wheeler, and D. Carruth, "Mississippi state university autonomous vehicle simulation library," in *Proc. Ground Vehicle Syst. Eng. Technol. Symp.*, 2020, pp. 11–13.
- [34] Z. Zainal and W. Rahiman, "Effect of yaw rate and sideslip to lateral vehicle dynamic stability," in *Proc. 10th Int. Conf. Robot., Vis., Signal Process. Power Appl.*, Berlin, Germany: Springer, 2019, pp. 473–479.
- [35] J. Y. Wong, *Terramechanics and Off-Road Vehicle Engineering: Terrain Behaviour, Off-Road Vehicle Performance and Design*. London, U.K.: Butterworth, 2009.
- [36] D. Newman, "The distribution of range in samples from a normal population, expressed in terms of an independent estimate of standard deviation," *Biometrika*, vol. 31, nos. 1–2, p. 20, Jul. 1939.



Siyuan Yu received the B.S.E. degree in electrical and computer engineering from Shanghai Jiao Tong University, China, in 2020, and the B.S.E. and M.S. degrees in mechanical engineering from the University of Michigan, Ann Arbor, in 2020 and 2022, respectively, where he is currently pursuing the Ph.D. degree. His research interests include modeling, system identification, motion planning, and control, with respect to vehicle systems.



Congkai Shen received the Bachelor of Science degree in electrical and computer engineering from Shanghai Jiao Tong University, China, in 2020, and the B.S. and M.S. degrees in mechanical engineering from the University of Michigan, Ann Arbor, in 2020 and 2022, respectively, where he is currently pursuing the Ph.D. degree. His research focuses on modeling, system identification, motion planning, and control, in the context of vehicle systems.



James Dallas received the B.S. degree in mechanical engineering from The Pennsylvania State University, University Park, PA, USA, in 2017, and the M.S. and Ph.D. degrees in mechanical engineering from the University of Michigan, Ann Arbor, MI, USA, in 2018 and 2021, respectively. His research interests include modeling, system identification, system dynamics and control, shared control, and adaptive and robust optimal control, with applications to vehicle systems.



Paramsothy Jayakumar received the M.S. and Ph.D. degrees in structural dynamics from Caltech and the B.S.E. degree from the University of Peradeniya, Sri Lanka. He is a Senior Technical Expert in modeling and simulation, SAE Fellow, and a member of the Analytics Team at U.S. Army Ground Vehicle Systems Center, Warren, MI. He is a member of U.S. Army Acquisition Corps, an Honorary Fellow of the Department of Mechanical Engineering, University of Wisconsin Madison, an Associate Editor of the ASME *Journal of Computational and Nonlinear Dynamics*, and an Editorial Board Member of the *International Journal of Vehicle Performance* and *Journal of Terramechanics*.



Bogdan I. Epureanu received the Ph.D. degree in mechanical engineering from Duke University in 1999. He is currently a Roger L. McCarthy Professor and an Arthur F. Thurnau Professor at the Department of Mechanical Engineering, University of Michigan, and has a courtesy appointment in electrical engineering and computer science. He is the Director of the Automotive Research Center. His research focuses on nonlinear dynamics of complex systems, such as teaming of autonomous vehicles, enhanced aircraft safety and performance, aeromechanics, early detection of neurodegenerative diseases, forecasting tipping points in disease epidemics, and ecology.

chanics, early detection of neurodegenerative diseases, forecasting tipping points in disease epidemics, and ecology.



Tulga Ersal received the B.S. degree in mechanical engineering from the Istanbul Technical University, Istanbul, Türkiye, in 2001, and the M.S. and Ph.D. degrees in mechanical engineering from the University of Michigan, Ann Arbor, MI, USA, in 2003 and 2007, respectively, where he is currently an Associate Professor with the Department of Mechanical Engineering. His research interests include modeling, simulation, and control of dynamic systems, with applications to vehicle and energy systems.